

# SUMIT YADAV

Mechanistic Interpretability | AI Alignment & Safety | NLP

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## RESEARCH STATEMENT

I work at the intersection of **mechanistic interpretability**, **AI alignment**, and **robustness**. My research studies how safety-aligned LLMs fail silently — through over-refusals, geometric misrepresentation, and surface-level triggers — and how the internal structure of representations can be understood and steered to make models safer and more reliable. Much of my strongest work is sole-authored: I combine representation-space analysis, adversarial probing, and systems engineering into empirically grounded methods. My work has been accepted at **ACL 2026 (CORE A\*)**, and I am especially interested in extending interpretability and safety to multilingual and low-resource settings.

## SELECTED RESEARCH & PUBLICATIONS

**SafeConstellations: Mitigating Over-Refusals in LLMs Through Task-Aware Representation Steering** | *ACL 2026 Main (CORE A\*)* 2026

- LLMs trace consistent, task-specific “constellation” trajectories in embedding space that shift predictably between refusal and non-refusal states.
- **SafeConstellations** is an inference-time, task-aware trajectory-shifting method that nudges representations toward non-refusal pathways *only* on over-refusal-prone tasks — cutting over-refusals by up to **73%** with minimal utility loss and no retraining. (with U. Maskey, M. Dras, U. Naseem.)

**Geometric Phases of Mechanism Formation in Neural Networks** (sole-authored) | *GLOW @ IJCAI-ECAI 2026 (Best Paper Candidate)* 2026

- Using linear probes and CKA across dense training checkpoints, shows classification mechanisms form **output-layer-first** and within the first ~5% of training (Cohen’s  $d = 3.68$ ); the deep-first pattern holds in from-scratch LLM pretraining (GPT-2 Small, SmoLM2-135M) and reproduces on public Pythia and OLMo-2 checkpoints.

**On the Relationship Between Representation Geometry and Generalization** (sole-authored) | *Pre-Print 2026* 2026

- Establishes that a label-free geometric metric — **effective dimension** — strongly predicts generalization across vision and language (partial  $r = 0.75$  over 52 ImageNet models, 13 architectures; carries to NLP encoders and decoder-only LLMs). The link is causal: degrading the geometry hurts accuracy while PCA cleanup preserves it — a diagnostic relevant to distribution-shift detection and robustness.

**Steering Hindi Competence into Unsupported Devanagari Languages** | *working paper* 2026

- With **no fine-tuning**, steers a model’s Hindi representations to elicit related Devanagari languages it was never trained on. On Llama-3.1-8B and Aya-23-8B a single steering direction produces **genuine Nepali**; Hindi outperforms English as the source because it sits closer in representation space, not because there is more of it. Unifies the SafeConstellations and maiBERT lines of work.

**The Steerability Spectrum of Visual Attributes** | *ongoing* 2026

- Measures how readily different visual features (size, distance, orientation) can be steered inside vision models (DINOv2, CLIP, SigLIP): some lie on a single clean, controllable direction while others are distributed and resist steering — probing what accounts for the difference.

**MaiBERT: A Pre-training Corpus and Language Model for Low-Resourced Maithili** | *LoResLM @ ACL 2026* 2026

- First monolingual BERT for Maithili (~50M speakers); **87.02%** news-classification accuracy, outperforming multilingual baselines (MuRIL, NepBERTa) — addressing fairness and access for underrepresented communities.

Also: **Revolutionizing Currency Security with YOLOv8** (J. Bus. Econ. Stud., 2024; 0.986 true-positive recall) · **Machine Learning Analysis of Tirhuta Lipi** (2023) · **Support Vectors for Imbalanced Text Classification** (2023).

## EXPERIENCE

**Astha.ai** USA (Remote)  
*AI Researcher — Safety & Agentic Systems* May 2025 – Present

- Built Zero-Trust agent oversight (cryptographic identity + policy engine on every tool call), an MCP security proxy (hexagonal architecture), and **MCP-Scanner** — a vulnerability platform integrating **78+** MITRE ATT&CK-mapped attack techniques for AI-agent infrastructure.

**AMNIL Technologies** Nepal (Hybrid)  
*AI Engineer — RAG & Infrastructure* Jun 2024 – May 2025

- Guardrails + **LLM-as-a-Judge** evaluation; hybrid-search RAG (Qdrant, +35% retrieval accuracy); self-hosted LLM serving with vLLM (40% lower latency).

**GradeUp Educations** Kathmandu, Nepal  
*Data Team Lead* Jan 2022 – Jun 2024

- Led the data team: learning agents/chatbots, an automated grade-evaluation system, and semantic-similarity matching (NLP + image processing).

**DeepLearning.AI** Global (Remote)  
*GAN Specialization Mentor* Aug 2021 – Present

- Technical mentor for the GANs Specialization (loss functions, training stability, generative-model robustness).

## SELECTED PROJECTS & OPEN SOURCE

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**Removing the “Workspace” Layers of Llama** — interpretability study locating a decodable-yet-entangled middle band of layers in Llama-3.1-8B; ablating the whole band collapses reasoning while shallow recall survives (linear probes + reversible layer-ablation, nnsight/NDIF to 405B). [code].

**World Models (toy experiments)** — built Dreamer-style world models (RSSM: GRU + stochastic latents) from scratch for continuous and visual control; agents trained purely in imagination transfer to the real task (CartPole 500/500), and short-horizon dreams with periodic reality-grounding cut drift  $11\times$ . [code].

**SAFE-MCP** — community security framework for the Model Context Protocol (ATT&CK-style threat/defense rules; prompt-injection & tool-poisoning modules).

**Telebot-Claude-Bridge** — permission-gated, human-in-the-loop remote control of the Claude Code CLI (inline allow/deny per tool call, audit logging).

**Maithili speech** — open Whisper-tiny **ASR** and MMS-TTS **TTS** for Maithili (live HF Spaces demos); **maiBERT** — first open BERT for Maithili (HF).

**NPL Coders** — founder of Nepal’s national data-science competition community.

## TECHNICAL PROFICIENCY

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**Interpretability & Alignment:** representation & activation steering, mechanistic interpretability, linear probes / CKA, representation geometry, effective dimension, over-refusal mitigation.

**LLMs & NLP:** LLMs (Llama, Claude, GPT), fine-tuning (PEFT/LoRA), RAG, multilingual / low-resource NLP, tokenizer training, embedding alignment.

**Agentic & Security:** Model Context Protocol, multi-agent frameworks, MITRE ATT&CK mapping, prompt-injection & tool-poisoning defense, adversarial evaluation.

**Tools:** Python (expert), PyTorch, TensorFlow, vLLM, FastAPI, Docker, Kubernetes, vector DBs (Qdrant, ChromaDB), nnsight.

**Research methods:** empirical NLP evaluation, information-theoretic metrics, adversarial benchmarking, statistical analysis.

## EDUCATION

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**Pulchowk Engineering College, IOE — Tribhuvan University**

Kathmandu, Nepal

*Bachelor of Computer Engineering*

2019 – 2024

- B.E. thesis: “Evaluating Auto-Encoding Transformer Language Models for Maithili Text Classification” — my first Maithili masked language model, and the foundation for maiBERT.
- Key subjects: Artificial Intelligence, Data Structures & Algorithms, Distributed Systems, Big Data Technologies, Network Security, Compiler Design.

## HONORS & AWARDS

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**Best Paper Candidate, GLOW @ IJCAI-ECAI 2026** (sole-authored) · Outstanding Reviewer, Mechanistic

Interpretability Workshop (ICML 2026) · Winner, GritFeat AI Hackathon (2023) · 1st Runner-Up, Locus Dataverse (2023)

Best AI Project, DELTA 3.0 (2023) · Winner, IT-Meet Image Challenge (2022) · Winner, LogPoint Capture The Flag (2022)